Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions to guide strategic business decisions.

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Dataset at a Glance

Dataset Summary

- **3,900** transactions across **18** columns
- Customer demographics, purchase details, and shopping behavior
- **37** missing values in Review Rating — imputed using median rating per product category

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Data Preparation in Python

Clean & Standardize

Loaded data with pandas, checked structure via df.info(), renamed columns to snake_case for consistency

Feature Engineering

Created age_group by binning ages and purchase_frequency_days from purchase data

Data Consistency

Dropped redundant promo_code_used column; verified discount_applied coverage

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL-based business analysis

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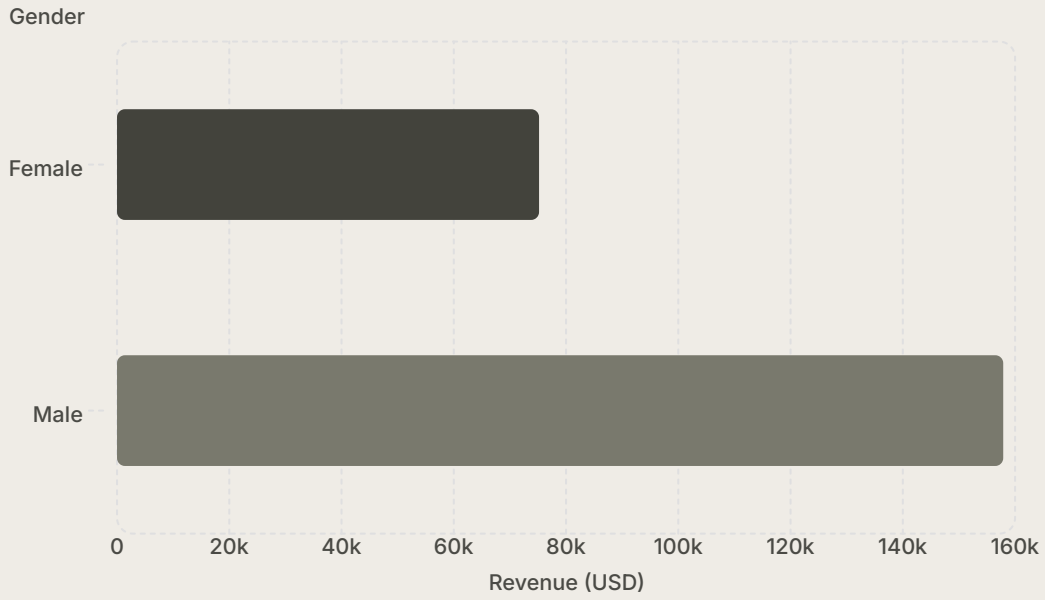
Revenue by Gender



Key Insight

Male customers generated **$157,890** in total revenue — more than double the **$75,191** from female customers. This significant gap highlights a key opportunity for targeted campaigns to increase female customer engagement and spending.

Male customers represent the dominant revenue driver in this dataset.

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Products & Shipping Insights

Top 5 Products by Review Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Shipping Type: Avg Purchase Amount

Shipping Type	Avg Amount
Standard	$58.46
Express	$60.48

Express shipping customers spend slightly more on average — a signal for upsell opportunities.

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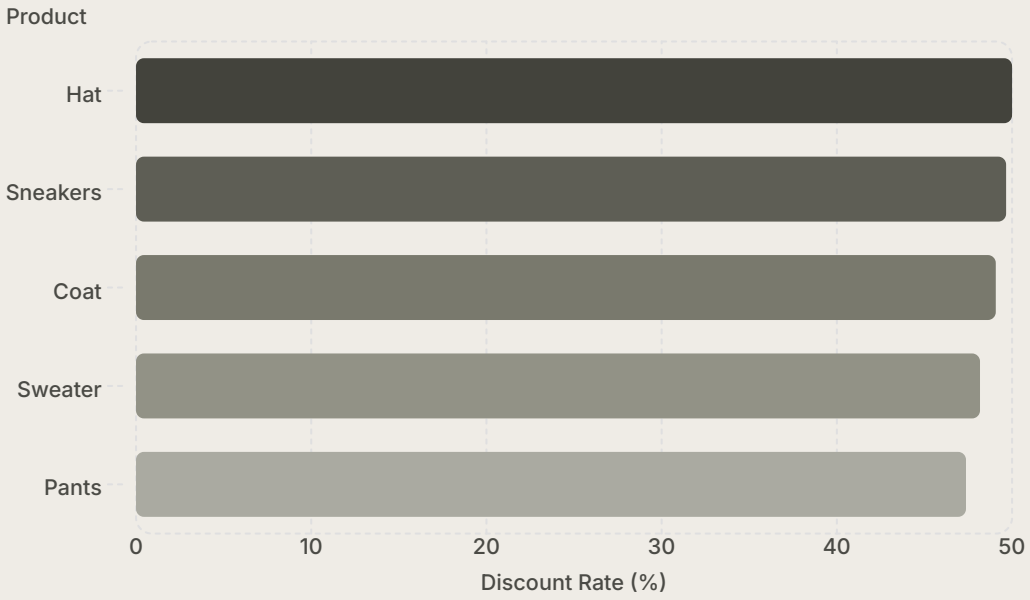
Discount Behavior & Subscriptions



Discount-Dependent Products

Hats, Sneakers, and Coats have the highest discount rates — nearly **50%** of purchases use a discount.

Subscribers vs. Non-Subscribers

Status	Customers	Revenue
Yes	1,053	$62,645
No	2,847	$170,436

Non-subscribers outnumber subscribers nearly **3:1** and drive the majority of revenue.

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Customer Segmentation

3,116

Loyal Customers

Highest purchase frequency — the core revenue base

701

Returning

Moderate repeat buyers with growth potential

83

New Customers

First-time buyers — prime targets for retention campaigns

Customers with **5+ purchases**: 958 subscribed vs. 2,518 did not — repeat buyers are more likely to subscribe.

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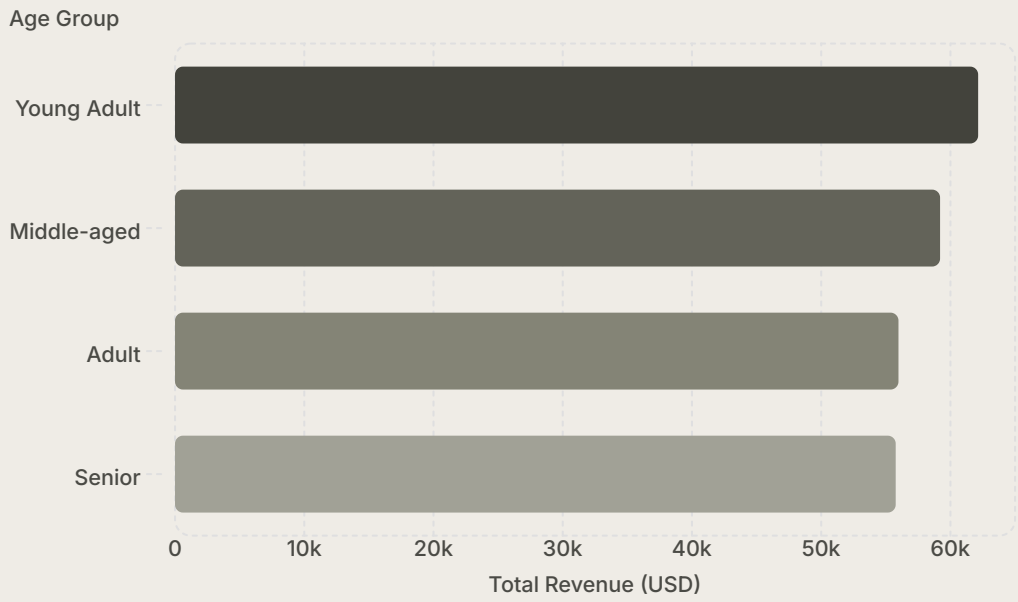
Revenue by Age Group



Age Group Insights

Young Adults contribute the highest total revenue at **$62,143**, closely followed by Middle-aged customers at **$59,197**. Revenue declines gradually across Adult and Senior segments, suggesting younger demographics are the primary purchasing drivers.

Focus marketing efforts on Young Adult and Middle-aged segments for maximum revenue impact.

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Power BI Dashboard

Dashboard Highlights

- **KPIs:** 3.9K customers · $59.76 avg purchase · 3.75 avg rating
- **Filters:** Subscription Status, Gender, Category, Shipping Type
- **Visuals:** Revenue & Sales by Category and Age Group, Subscription breakdown (27% Yes / 73% No)

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Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base

→ Customer Loyalty Programs

Reward repeat buyers to grow the Loyal segment (currently 3,116 customers)

→ Review Discount Policy

Balance sales boosts with margin control — nearly 50% of top products are discount-dependent

→ Product Positioning

Highlight top-rated items (Gloves, Sandals, Boots) in marketing campaigns

→ Targeted Marketing

Focus on high-revenue age groups and express-shipping users for maximum ROI

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